STUDENT REFERENCE & INTERNAL USE NOTICE

This computational workflow documents reproducible 16S rRNA gene amplicon sequence analysis using QIIME 2 and DADA2 in HOPE Lab research and is PROVIDED ONLY FOR THE LAB MEMBERS for training and internal reference. It serves as a reproducible command template; users must verify compatibility with their installed QIIME 2 distribution, plugin versions, sequencing design, and institutional data-governance policies. External users should independently assess method suitability and obtain required approvals, if any. This document does not replace an institutionally approved SOP or method-specific risk assessment.

TITLE

16S rRNA Gene Amplicon Sequence Analysis Using QIIME 2 and DADA2 - Computational Workflow

PRINCIPLE

16S rRNA gene amplicon sequencing characterizes bacterial community structure and diversity by analyzing DNA sequence reads from targeted hypervariable regions (e.g., V3–V4 or V4) of the bacterial 16S ribosomal RNA gene. Starting from demultiplexed paired-end FASTQ files, a reproducible bioinformatic pipeline validates sample-to-file mapping, imports sequence data, inspects base quality profiles, removes residual primer/adapter sequences, denoises reads into exact amplicon sequence variants (ASVs), assigns taxonomic classifications, and computes alpha- and beta-diversity metrics.

Unlike legacy operational taxonomic unit (OTU) clustering at an arbitrary 97% similarity threshold, DADA2 applies a parametric error model to resolve single-nucleotide differences and infer exact ASVs (Callahan et al., 2016). Analytical reliability depends directly on read quality filtering, primer trimming, sufficient forward/reverse overlap for paired-end merging, region-matched taxonomic classifiers, negative-control evaluation, and complete software provenance tracking.

Learning Outcomes

- Validate sample metadata tables and paired-end FASTQ manifest files prior to data import.
- Import paired-end sequence reads into QIIME 2 artefacts and evaluate per-base quality score distributions.
- Select data-driven trimming and truncation lengths based on quality decay and minimum amplicon overlap.
- Execute DADA2 denoising and audit per-sample read retention across filtering, merging, and chimera removal.
- Assign taxonomy using a region-matched classifier, construct phylogenetic trees, and evaluate negative controls.

OBJECTIVE & SCOPE

Objective: To provide a standardized, reproducible computational workflow for inferring ASVs and characterizing bacterial taxonomic composition and community diversity from 16S rRNA amplicon sequencing data.

Scope: Covers dry-lab bioinformatic processing beginning with demultiplexed digital FASTQ files through diversity analysis and provenance archiving. Excludes wet-lab sample collection, DNA extraction, PCR amplification, and

sequencing. Amplicon sequencing does not assess cell viability, guarantee species/strain-level resolution, or directly measure functional gene abundance.

CORE BIOINFORMATIC CONCEPTS

- **Amplicon Sequence Variants (ASVs):** DADA2 infers exact biological sequences prior to amplification and sequencing errors, offering higher resolution and cross-study reproducibility than 97% OTU clustering.
- **Paired-End Overlap Requirement:** Retained forward and reverse read lengths after primer trimming and quality truncation must exceed the target amplicon length by at least the minimum overlap threshold (typically ≥12 nt in DADA2) to merge reliably.
- **Region-Matched Taxonomic Classifiers:** Naive Bayes classifiers trained specifically on the amplified hypervariable region and primer pair yield substantially higher classification accuracy than full-length or mismatched classifiers.
- **Control-Informed Contamination Review:** Extraction blanks and PCR no-template controls (NTCs) capture reagent and cross-well background DNA. Evaluate prevalence and abundance patterns systematically rather than applying arbitrary blanket feature deletion.

DATA, SOFTWARE & COMPUTATIONAL REQUIREMENTS

Data Inputs & Reference Files

- Demultiplexed paired-end FASTQ files (.fastq.gz) for forward (R1) and reverse (R2) reads
- Tab-separated V2 manifest file (sample-id, forward-absolute-filepath, reverse-absolute-filepath)
- Validated sample metadata TSV file including all biological samples and negative controls
- Primer sequences, target hypervariable region details, and quality encoding (PHRED 33)
- Region-matched reference taxonomy database & pre-trained classifier artefact (.qza)

Software & Hardware Environment

- Supported QIIME 2 Amplicon Distribution (active Conda environment)
- Core plugins: q2-types, q2-demux, q2-cutadapt, q2-dada2, q2-feature-table, q2-feature-classifier, q2-phylogeny, q2-diversity
- Multi-core workstation/server (≥16–32 GB RAM recommended for DADA2 & taxonomic classification)
- Sufficient storage capacity for raw FASTQ files, .qza artefacts, .qzv visualisations, and provenance
- Optional R (phyloseq, vegan) or Python packages for downstream statistical modeling

STEP-BY-STEP COMPUTATIONAL WORKFLOW

Step 1: Record Software Environment & Validate Manifest/Metadata

Activate the installed QIIME 2 environment and archive complete version details using: qiime info > qiime2_environment_info.txt. Verify that the tab-separated manifest uses exact V2 headers (sample-id, forward-absolute-filepath, reverse-absolute-filepath), all file paths exist, and sample IDs match metadata.tsv exactly, including extraction blanks and PCR controls.

Step 2: Import Paired-End Reads & Inspect Quality Profiles

Import demultiplexed PHRED 33 reads and generate quality summaries:
qiime tools import --type 'SampleData[PairedEndSequencesWithQuality]' --input-path manifest.tsv --output-path demux.qza --input-format PairedEndFastqManifestPhred33V2
qiime demux summarize --i-data demux.qza --o-visualization demux-summary.qzv
Inspect forward and reverse quality plots in QIIME 2 View to determine read counts, read lengths, and base-quality drop-off positions.

Step 3: Review & Remove Residual Primer/Adapter Sequences

If PCR primers or sequencing adapters remain in the reads, trim them prior to denoising using q2-cutadapt (qiime cutadapt trim-paired) with documented forward and reverse primer sequences. Record the exact primer sequences, error tolerance, and read retention rates.

Step 4: Denoise Paired-End Reads with DADA2

Run DADA2 using trimming and truncation parameters justified by the quality profiles and expected amplicon overlap:

qiime dada2 denoise-paired --i-demultiplexed-seqs demux.qza --p-trim-left-f [TRIM_F] --p-trim-left-r [TRIM_R] --p-trunc-len-f [TRUNC_F] --p-trunc-len-r [TRUNC_R] --o-representative-sequences rep-seqs.qza --o-table table.qza --o-denoising-stats stats.qza

Inspect stats.qza via qiime metadata tabulate to verify read retention through filtering, denoising, merging, and non-chimeric steps.

Step 5: Summarise Feature Table & Assign Taxonomy

Generate feature summaries and classify ASVs using a classifier trained on the matched 16S region:

qiime feature-table summarize --i-table table.qza --m-sample-metadata-file metadata.tsv --o-visualization table-summary.qzv

qiime feature-classifier classify-sklearn --i-classifier [REGION_CLASSIFIER.qza] --i-reads rep-seqs.qza --o-classification taxonomy.qza

Report taxonomic assignments strictly at the resolution supported by the amplified marker region (typically genus level for short 16S amplicons).

Step 6: Construct Phylogenetic Tree & Compute Diversity Metrics

Build a rooted phylogenetic tree and compute core diversity metrics at a justified rarefaction depth:

qiime phylogeny align-to-tree-mafft-fasttree --i-sequences rep-seqs.qza --o-alignment aligned-rep-seqs.qza --o-masked-alignment masked-aligned-rep-seqs.qza --o-tree unrooted-tree.qza --o-rooted-tree rooted-tree.qza

qiime diversity core-metrics-phylogenetic --i-phylogeny rooted-tree.qza --i-table table.qza --p-sampling-depth [DEPTH] --m-metadata-file metadata.tsv --output-dir core-metrics

Step 7: Evaluate Negative Controls & Archive Provenance

Inspect read counts and ASV compositions in extraction blanks and PCR no-template controls. Apply a documented contamination-assessment strategy without arbitrary blanket feature removal. Archive raw FASTQ files, manifests, metadata, exact shell scripts, .qza/.qzv artefacts, and accession numbers.

CRITICAL FACTORS AFFECTING RESULTS

Variable	Effect of Inadequate Control	Required Control Standard
Manifest & Sample IDs	Mismatched file paths swap sample identities across dataset	Validate exact paths, unique IDs & R1/R2 pairing pre-import
Primer Removal	Residual degenerate primers inflate artificial ASV diversity	Trim primers via q2-cutadapt or verify upstream removal
Truncation Lengths	Under-truncation retains errors; over-truncation breaks overlap	Select lengths from quality profiles while preserving ≥12 nt overlap
Taxonomy Classifier	Mismatched 16S region/database lowers classification accuracy	Use current classifier trained on exact primer/amplicon region

Sampling Depth	Arbitrary rarefaction drops low-depth samples or loses richness	Justify depth via table-summary & alpha-rarefaction curves
Negative Controls	Reagent/kit contaminants misinterpreted as biological taxa	Sequence blanks/NTCs alongside samples; document filtering

QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Audit demux-summary.qzv, DADA2 per-sample denoising stats, feature-table summaries, and negative-control ASV profiles before statistical testing. Retain complete embedded QIIME 2 provenance in all .qza and .qzv files.

Problem / Observation	Likely Cause	Corrective Action
Import command fails	Manifest header typo, relative paths, or encoding mismatch	Verify V2 TSV header, absolute paths, and PHRED 33 encoding
Low read merging in DADA2	Truncation lengths too short for target amplicon overlap	Increase truncation lengths if quality permits; verify primer trim
High unassigned fraction	Classifier trained on wrong 16S region or reverse-complemented reads	Verify read orientation and match classifier to amplified region
Phylogenetic diversity error	Feature IDs in filtered table missing from rooted tree	Rebuild tree from filtered rep-seqs or filter tree consistently
Reads in negative controls	Background kitome DNA, low biomass, or index hopping	Compare control vs. sample prevalence; document decontamination

DATA MANAGEMENT & BIOSAFETY

- **Biosafety & Upstream Compliance:** Although computational analysis handles digital files, upstream nucleic-acid extraction and PCR must follow institutional BSL-1/BSL-2 biosafety protocols. Ensure ethical approvals and participant/host privacy protections are verified.
- **Data Governance & Archiving:** De-identify human or sensitive host metadata prior to public repository submission (e.g., NCBI SRA / ENA). Retain raw FASTQ files, metadata, environment logs, and exact command scripts according to institutional data-retention policies.

REFERENCES & RELATED DOCUMENTS

Hossain TJ, Hossain S, Nafiz IH, Islam R, Khan MS, Alhumaidan OS, et al. (2023). Metagenomic analysis of bacterial diversity and community in date palm sap: dominance of Leuconostoc, Zymomonas, and Lactobacillus. Preprint.

Hossain TJ. (2022). Functional genomics of the lactic acid bacterium Limosilactobacillus fermentum LAB-1: metabolic, probiotic and biotechnological perspectives. Heliyon, 8(9), e10704. doi:10.1016/j.heliyon.2022.e10704